

APPLICATIONS SECTION

BAYESIAN ESTIMATION IN MIXED LINEAR
MODELS USING THE GIBBS SAMPLER

A.J. v.d. Merwe: Department of Mathematical Statistics, University of
and the Orange Free State, P.O. Box 339, Bloemfontein,
T.J. Botha: 9300 Republic of South Africa

Key words: Classical estimators; credibility intervals; fixed effects;
Gibbs sampler; mixed linear models; posterior distributions;
variance components.

Summary: Mixed linear models are naturally modelled within the Bayesian framework. Nonetheless, calculation of the marginal posterior distributions of variance components and functions of variance components has proved a challenging technical problem. The purpose of this paper is to show that an adaptive Monte Carlo integration technique known as the Gibbs sampler provides a simple and efficient method for calculating marginal posterior distributions and expectations of fixed and random effects.

1. INTRODUCTION

Box and Tiao (1973) wrote: "The sampling theory approach to the problem of variance components encounters a number of snags. These have bothered statisticians for many years, as is evidenced by the great variety of attempts which have been made to resolve the problems."

The above statement applies particularly well to the estimation of variance components with mixed linear models and unbalanced data for which a number of methods have been developed during the last 30 years. For more information on these methods see for example Searle (1971, 1979) and Henderson (1984).

Estimation of variance components using Bayesian methods has been

explored by Hill (1965, 1977), Lindley and Smith (1972), Harville (1977), Gianola and Fernando (1986), Chaloner (1987), and Gianola and Foulley (1990). However, the technical problems encountered in attempting to carry out the required numerical integrations to obtain formally these marginal posterior distributions and expectations, have long served as an impediment to the wider application of the Bayesian framework to real data.

Box and Tiao (1973) reported a substantial amount of detailed, sophisticated approximation work; both analytic and numerical. Skene (1983) considered purpose-built numerical techniques. The methods described by Smith *et al.* (1985, 1987) require careful reparameterization dependent upon both the data and the choice of priors. In a similar spirit, Achcar and Smith (1990) discussed parameter transformations for successful implementation of Laplace's method (Tierney and Kadane (1986)).

The objective of the present paper is to present the Bayesian method of variance component estimation in mixed linear models based on the principle of finding the marginal posterior distributions of the unknown parameters. These distributions will be observed by using a sampling based approach known as the Gibbs sampler. By comparison with the earlier methods mentioned above the Gibbs sampler is remarkably simple.

In three recent papers (Gelfand and Smith (1990, 1991) and Gelfand *et al.* 1990)) the Gibbs sampler was discussed in the context of the calculation of marginal posterior densities for a wide range of Bayesian problems, several of which were previously intractable. The Gibbs sampler offers conceptual simplicity and straightforward implementation without sophisticated numerical or analytic approximation expertise and associated specialist software (as for example, in Naylor and Smith (1982, 1988), Smith *et al.* (1985, 1982), Shaw (1988) and Geweke (1988)). The Gibbs sampler has been implemented for very high dimensional problems (Geman and Geman (1984)), e.g. 10^5 or more variables. By comparison even a complicated Bayesian application would usually have low dimension.

We will briefly review the Gibbs sampler in section 2. In section 3 the Gibbs sampler is used to obtain marginal posterior distributions in a mixed

linear model. In addition, we illustrate the ease with which inferences for functions of parameters such as ratios, can be made. For the example given in section 3, the Bayesian results will be compared with the results obtained from conventional methods used by geneticists. In section 4 a comparison is made between numerical integration and the Gibbs sampler and in section 5 a sensitivity analysis is performed on some of the prior specifications of the model.

2. THE GIBBS SAMPLER

Gibbs sampling is a technique for generating random variables from a (marginal) distribution indirectly, without having to calculate the density. The mechanism is based only on elementary properties of Markov chains.

Consider a given joint density $f(x, y, z)$ and suppose we are interested in obtaining characteristics of the marginal density

$$f(x) = \int \int f(x, y, z) dy dz$$

such as the mean and variance. The most natural and straightforward approach would be to calculate $f(x)$ analytically or even numerically, and then use this result to obtain the desired characteristic. However, there are many cases where the marginal distribution $f(x)$ cannot be readily calculated.

The Gibbs sampler is a method for generating a sample from $f(x)$ in an indirect way, by sampling from the conditional distributions $f(x|y, z)$, $f(y|x, z)$ and $f(z|x, y)$ which are often known in linear statistical models. It must therefore be possible to generate samples from these conditional distributions given the specified values of the conditioning variables.

The algorithm also requires that the joint density $f(x, y, z)$ is uniquely determined by the conditional densities $f(x|y, z)$, $f(y|x, z)$ and $f(z|x, y)$.

The Gibbs sampler generates a random sample

$$X_1, \dots, X_m \sim f(x)$$

indirectly as follows: Consider the arbitrary starting values, $y^{(0)}$ and $z^{(0)}$.
Draw

$$\begin{aligned} X^{(1)} & \text{ from } f(x|y^{(0)}, z^{(0)}); \\ \text{then } Y^{(1)} & \text{ from } f(y|x^{(1)}, z^{(0)}) \\ \text{and } Z^{(1)} & \text{ from } f(z|x^{(1)}, y^{(1)}) \end{aligned}$$

to complete one iteration of the scheme. At the k -th iteration we draw

$$\begin{aligned} X^{(k)} & \text{ from } f(x|y^{(k-1)}, z^{(k-1)}); \\ \text{then } Y^{(k)} & \text{ from } f(y|x^{(k)}, z^{(k-1)}) \\ \text{and } Z^{(k)} & \text{ from } f(z|x^{(k)}, y^{(k)}). \end{aligned}$$

Geman and Geman (1984) have shown that under fairly general conditions the distribution of $X^{(k)}$ converges to $f(x)$ for large enough k . The value $X^{(k)}$ can thus be regarded as a simulated observation from $f(x)$. By repeating the process m times the Gibbs sampler generates m observations $X_1^{(k)}, \dots, X_m^{(k)}$. These m observations can now be regarded as an independent sample from $f(x)$ on condition that the iterations are independent using predetermined starting values $y^{(0)}$ and $z^{(0)}$ for each iteration. By simulating a large enough sample, the mean, variance or any other characteristic of $f(x)$ can be calculated to the desired degree of accuracy. Characteristics of $f(y)$ and $f(z)$ can be obtained in a similar way.

For further details about the Gibbs sampler see, for example, Casella and George (1990), Gelfand and Smith (1990, 1991) and Gelfand *et al.* (1991).

3. A MIXED LINEAR MODEL

The data used for this illustration are a subset of those used by Harvey (1975) and are given in table 3.1. The data were also analysed by Kennedy (1981) who compared several classical approaches to the joint estimation of

variance components and prediction of breeding values. Kennedy also discussed some of the unresolved problems in this area. In table 3.1 average daily gain (ADG) is measured on 36 steers of the Hereford breed. The steers are classified as to line (fixed), age of dam (fixed) and sire within line (random).

TABLE 3.1: Data on average daily gain of steers

Steer	Line	Sire	Age of dam	Average daily gain	Steer	Line	Sire	Age of dam	Average daily gain
1	1	1	3	2,24	19	1	3	5+	2,29
2	1	1	3	2,65	20	1	3	5+	2,30
3	1	1	4	2,41	21	1	3	5+	2,94
4	1	1	4	2,25	22	2	4	3	2,50
5	1	1	5+	2,58	23	2	4	3	2,44
6	1	1	5+	2,67	24	2	4	4	2,44
7	1	1	5+	2,71	25	2	4	4	2,15
8	1	1	5+	2,47	26	2	4	5+	2,54
9	1	2	4	2,29	27	2	4	5+	2,74
10	1	2	4	2,26	28	2	4	5+	2,50
11	1	2	5+	1,97	29	2	4	5+	2,54
12	1	2	5+	2,14	30	2	5	3	2,65
13	1	2	5+	2,44	31	2	5	4	2,52
14	1	2	5+	2,52	32	2	5	4	2,67
15	1	2	5+	1,72	33	2	5	5+	2,79
16	1	2	5+	2,75	34	2	5	5+	2,33
17	1	3	3	2,38	35	2	5	5+	2,67
18	1	3	4	2,46	36	2	5	5+	2,69

In matrix notation the resulting mixed linear model can be written as

$$(3.1) \quad y = X_1\beta_1 + X_2\beta_2 + X_3\beta_3 + Z\underline{u} + \underline{\varepsilon}$$

where

$\underline{y}(n \times 1)$ = the vector of average daily gains - the data vector,

$\underline{\beta}_1(p_1 \times 1) = \mu$, the population mean,

$\underline{\beta}_2(p_2 \times 1) = [a_1 \ a_2 \ a_3]'$ where a_i is the fixed effect of the i -th age of dam,

$\underline{\beta}_3(p_3 \times 1) = [g_1 \ g_2]'$ where g_j is the fixed effect of the j -th breeding group or line.

$\underline{u}(q \times 1) = [s_{11} \ s_{12} \ s_{13} \ s_{21} \ s_{22}]'$ where s_{jk} is the random effect of the k -th sire within the j -th line.

In table 3.1, S_{21} is sire no. 4 and S_{22} sire no 5.

$\underline{\epsilon}(n \times 1)$ = the vector of random errors.

$X_i(n \times p_i)$ and $Z(n \times q)$ are known design matrices. For our example it follows that $n = 36$, $p_1 = 1$, $p_2 = 3$, $p_3 = 2$ and $q = 5$. It is also assumed that $\underline{u}$ and $\underline{\epsilon}$ are independently normally distributed:

$$\underline{u} \sim N(\underline{0}, \sigma_s^2 I_q) \text{ and } \underline{\epsilon} \sim N(\underline{0}, \sigma_e^2 I_n)$$

where I_q and I_n are the identity matrices of orders q and n respectively. σ_s^2 and σ_e^2 are the unknown sire and error components of variance. The idea is to use Bayesian techniques and Gibbs sampling to estimate σ_s^2 , σ_e^2 , $\underline{\beta}_1$, $\underline{\beta}_2$, $\underline{\beta}_3$, and $\underline{u}$.

The distribution

$$(3.2) \ f(\underline{y}, \underline{u} | \underline{\beta}_1, \underline{\beta}_2, \underline{\beta}_3, \sigma_s^2, \sigma_e^2) = f(\underline{y} | \underline{\beta}_1, \underline{\beta}_2, \underline{\beta}_3, \underline{u}, \sigma_e^2) f(\underline{u} | \sigma_s^2)$$

where

$$\underline{y} | \underline{\beta}_1, \underline{\beta}_2, \underline{\beta}_3, \underline{u}, \sigma_e^2 \sim N \left[\sum_{i=1}^3 X_i \underline{\beta}_i + Z \underline{u}, \sigma_e^2 I_n \right]$$

gives rise to the "full" likelihood function from which maximum likelihood estimators of $\underline{\beta}_1$, $\underline{\beta}_2$, $\underline{\beta}_3$, and each of the variance components can be obtained. In the Bayesian treatment of model (3.1) prior subjective uncertainty about $\underline{\beta}_1$, $\underline{\beta}_2$, $\underline{\beta}_3$ is often described by means of the "flat" prior density.

$$(3.3) \quad p(\underline{\beta}_1, \underline{\beta}_2, \underline{\beta}_3) \propto \text{constant}$$

and the normal distribution assigned to $\underline{y}$ is viewed as a prior probability distribution as well. The non-informative priors for the variance components are

$$(3.4) \quad p(\sigma_e^2, \sigma_s^2) = p(\sigma_e^2)p(\sigma_s^2) \propto \sigma_e^{-2} \sigma_s^{-2} \quad \begin{matrix} 0 < \sigma_e^2 < \infty \\ 0 < \sigma_s^2 < \infty \end{matrix}$$

(For further details about prior information, see for example Zellner (1971), Gianola and Fernando (1986) and Chaloner (1987).

From (3.2), (3.3) and (3.4) it follows that the joint density function can be written as

$$(3.5) \quad f(\underline{y}, \underline{\beta}_1, \underline{\beta}_2, \underline{\beta}_3, \underline{u}, \sigma_e^2, \sigma_s^2) = f(\underline{y}, \underline{u} | \underline{\beta}_1, \underline{\beta}_2, \underline{\beta}_3, \sigma_e^2, \sigma_s^2) p(\underline{\beta}_1, \underline{\beta}_2, \underline{\beta}_3) p(\sigma_e^2, \sigma_s^2) \\ \propto (\sigma_e^2)^{-\frac{1}{2}n_e} e^{-\frac{1}{2}\left\{\underline{y} - \sum_{i=1}^3 X_i \underline{\beta}_i - Z\underline{u}\right\}' \left\{\underline{y} - \sum_{i=1}^3 X_i \underline{\beta}_i - Z\underline{u}\right\} / \sigma_e^2} (\sigma_s^2)^{-\frac{1}{2}n_s} e^{-\frac{1}{2}\underline{u}' \underline{u} / \sigma_s^2} \sigma_e^{-2} \sigma_s^{-2}$$

The Gibbs sampler involves σ_e^2 , σ_s^2 , $\underline{\beta}_1$, $\underline{\beta}_2$, $\underline{\beta}_3$ and $\underline{u}$.

From (3.5) it follows that the required full conditionals are

$$(3.6) \quad \sigma_e^2 | \underline{y}, \underline{\beta}_1, \underline{\beta}_2, \underline{\beta}_3, \underline{u}, \sigma_s^2 \sim IG \left[\frac{1}{2}n, \frac{1}{2} \left\{ \underline{y} - \sum_{i=1}^3 X_i \underline{\beta}_i - Z\underline{u} \right\}' \left\{ \underline{y} - \sum_{i=1}^3 X_i \underline{\beta}_i - Z\underline{u} \right\} \right]$$

$$(3.7) \quad \sigma_s^2 | \underline{y}, \underline{\beta}_1, \underline{\beta}_2, \underline{\beta}_3, \underline{u}, \sigma_e^2 \sim IG \left(\frac{1}{2}q, \frac{1}{2}\underline{u}'\underline{u} \right)$$

$$(3.8) \quad \underline{u} | \underline{y}, \underline{\beta}_1, \underline{\beta}_2, \underline{\beta}_3, \sigma_e^2, \sigma_s^2 \sim N(D^{-1}Z'\underline{\theta}/\sigma_e^2, D^{-1})$$

and

$$(3.9) \quad \underline{\beta}_i | \underline{y}, \underline{u}, \underline{\beta}_{(-i)}, \sigma_e^2, \sigma_s^2 \sim N((X_i'X_i)^{-1}X_i'\underline{\theta}_{\beta_i}, \sigma_e^2(X_i'X_i)^{-1})$$

where

$$(3.10) \quad D = \frac{1}{\sigma_e^2} Z'Z + \frac{1}{\sigma_s^2} I_q, \quad \underline{\theta} = \underline{y} - \sum_{i=1}^3 X_i \underline{\beta}_i,$$

$$\underline{\theta}_{\beta_i} = \underline{\theta} - X_i \underline{\beta}_i - Z\underline{u} \quad \text{and} \quad i = 1, 2, 3.$$

$IG(a, b)$ denotes the Inverse Gamma distribution (i.e. for the variable σ^2

the density is of the form $\frac{b^a e^{-b/\sigma^2}}{\Gamma(a)(\sigma^2)^{a+1}}$).

The Gibbs sampling procedure can now easily be implemented. Standard routines will be used to generate random numbers from the required distributions. The iterative process starts by using arbitrary starting values $\underline{\beta}_1^{(0)}, \underline{\beta}_2^{(0)}, \underline{\beta}_3^{(0)}, \underline{u}^{(0)}$ in (3.6) to draw $\sigma_e^{2(1)}$ and also for (3.7) to draw $\sigma_s^{2(1)}$.

To complete the first iteration these values are put into (3.8) and (3.9) to draw $\underline{u}^{(1)}$; $\underline{\beta}_1^{(1)}, \underline{\beta}_2^{(1)}$ and $\underline{\beta}_3^{(1)}$. After k iterations in which the

conditional distributions were updated at each iteration, the Gibbs sampler has generated the values $\beta_1^{(k)}, \beta_2^{(k)}, \beta_3^{(k)}, \underline{y}^{(k)}, \sigma_s^2{}^{(k)}$ and $\sigma_e^2{}^{(k)}$. The process is then repeated m times. The results for this experiment have been obtained by using $k = 20$ and $m = 1\,000$. The convergence of the algorithm is discussed in section 5 where the posterior distributions of the unknown parameters are compared for various k . For bigger data sets and awkward shape posterior distributions larger k and m would be preferable. A few test runs with a specific data set, changing the values of k and m each time will indicate the optimum values.

Using the posterior conditional densities for σ_e^2 and σ_s^2 , the marginal posterior densities were estimated as

$$\hat{f}(\sigma_e^2 | \underline{y}) = \frac{1}{m} \sum_{i=1}^m f(\sigma_e^2 | \underline{y}, \beta_{1i}^{(k)}, \beta_{2i}^{(k)}, \beta_{3i}^{(k)}, \underline{y}_i^{(k)})$$

and

$$(3.11) \quad \hat{f}(\sigma_s^2 | \underline{y}) = \frac{1}{m} \sum_{i=1}^m f(\sigma_s^2 | \underline{y}, \underline{y}_i^{(k)})$$

and are displayed in figures 3.1 and 3.2.

With regard to point estimates of σ_e^2 and σ_s^2 the posterior means, medians or modes can be used. The marginal posterior means were estimated as

$$\hat{\sigma}_e^2 = \frac{1}{m} \sum_{i=1}^m E(\sigma_e^2 | \underline{y}, \beta_{1i}^{(k)}, \beta_{2i}^{(k)}, \beta_{3i}^{(k)}, \underline{y}_i^{(k)})$$

and

$$(3.12) \quad \hat{\sigma}_s^2 = \frac{1}{m} \sum_{i=1}^m E(\sigma_s^2 | \underline{y}, \underline{y}_i^{(k)}).$$

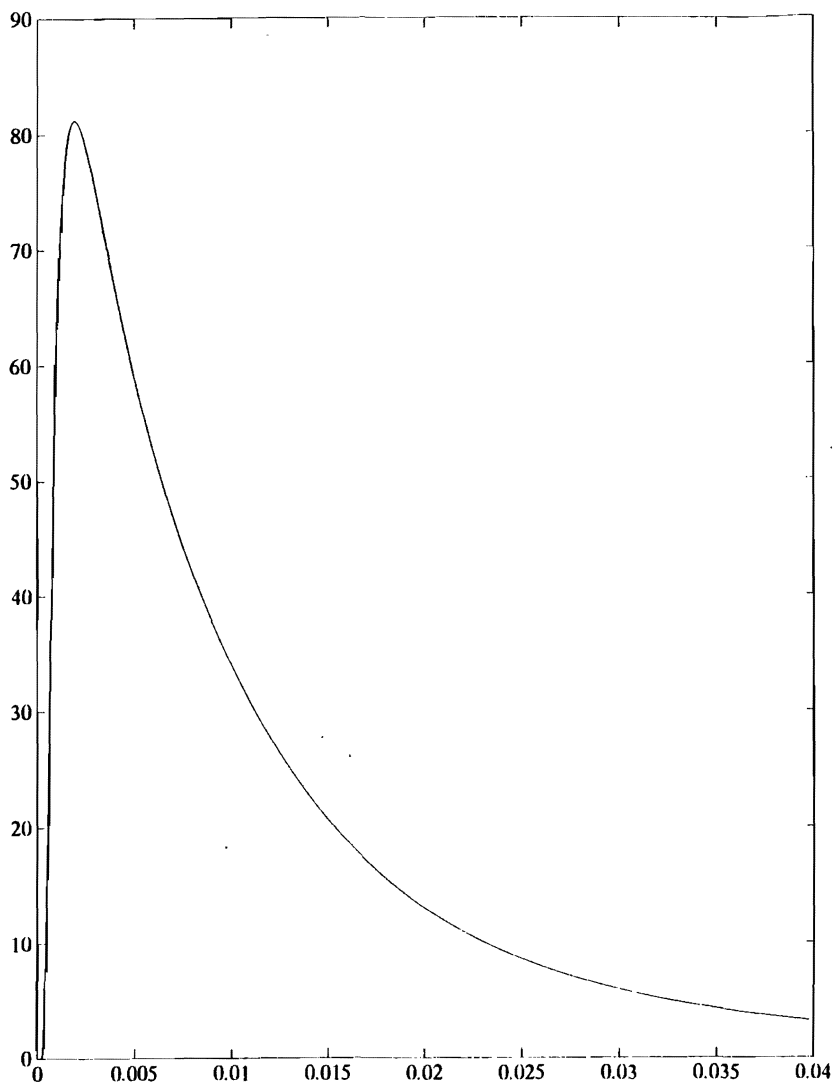


FIGURE 3.1: Posterior marginal for σ_s^2

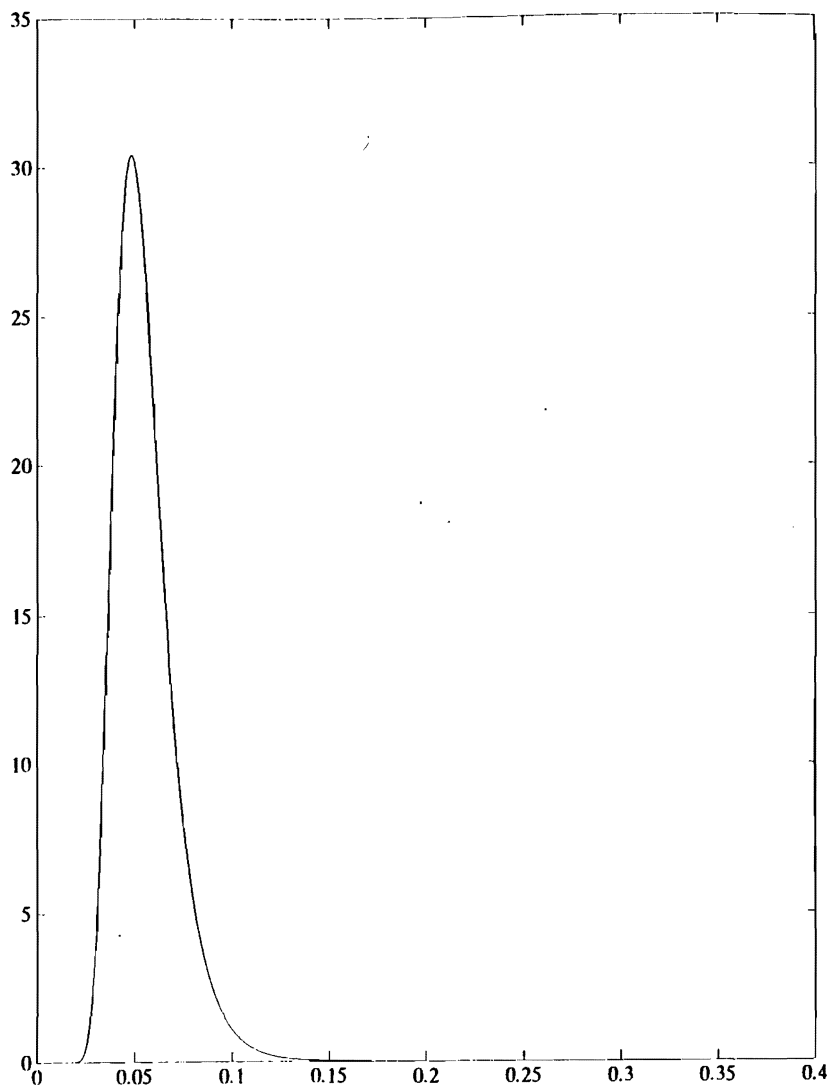


FIGURE 3.2: Posterior marginal for σ_e^2

These Rao-Blackwell estimates are more accurate than the usual point estimates $\hat{\sigma}_e^2 = \frac{1}{m} \sum_{i=1}^m \sigma_{ei}^{2(k)}$ and $\hat{\sigma}_s^2 = \frac{1}{m} \sum_{i=1}^m \sigma_{si}^{2(k)}$. For more details about the accuracy of these estimates see Gelfand and Smith (1991) and Casella and George (1990).

It is well known (and can be observed from figures 3.1 and 3.2) that the distributions of variance components and functions of variance components are often quite skew which means that the median might be a better estimator than the mean because it is less sensitive to extreme values (see Rubin (1986)). To estimate heritability we can use either

$$(3.13) \quad \hat{h}^2 = \frac{1}{m} \sum_{i=1}^m \frac{4\sigma_{si}^{2(k)}}{\sigma_{si}^{2(k)} + \sigma_{ei}^{2(k)}}$$

or the median. A Rao-Blackwell estimator for h does not exist. The

variance ratio $\frac{\sigma_e^2}{\sigma_s^2}$ is often a quantity of interest. Estimating $E \left[\frac{\sigma_e^2}{\sigma_s^2} | \underline{y} \right]$ is

easily accomplished by using one of the three methods:

$$(3.14) \quad \frac{1}{m} \sum_{i=1}^m \frac{1}{\sigma_{si}^2} E \left[\sigma_e^2 | \underline{y}, \underline{u}_i^{(k)} \right]$$

$$(3.15) \quad \frac{1}{m} \sum_{i=1}^m \sigma_{ei}^2 E \left[\frac{1}{\sigma_{si}^2} | \underline{y}, \underline{\beta}_{1i}^{(k)}, \underline{\beta}_{2i}^{(k)}, \underline{\beta}_{3i}^{(k)}, \underline{u}_i^{(k)} \right]$$

and

$$(3.16) \quad \frac{1}{m} \sum_{i=1}^m \frac{\sigma_{ei}^{2(k)}}{\sigma_{si}^{2(k)}}$$

Equations (3.14) and (3.15) use conditional expectations and are therefore Rao-Blackwell estimates.

In table 3.2 results from Henderson's method 3, the maximum likelihood estimator (ML), the restricted maximum likelihood estimator (REML), the minimum norm quadratic unbiased estimator (MINQUE) and some variations of MINQUE as calculated by Kennedy (1981) are compared with the Bayes estimators (median of the marginal posterior distributions) obtained from the Gibbs sampler. From table 3.2 it follows that there are only small differences between the Bayes estimators and the classical estimators of variance components.

TABLE 3.2: Estimates of variance components and heritability by various methods

	Method 3	ML	REML	MINQUE 1	MIQUEO	MINQUED 1	Gibbs Median
$\hat{\sigma}_s^2$	0,0096	0,0033	0,0095	0,0096	0,0109	0,0099	0,0090
$\hat{\sigma}_e^2$	0,0508	0,0474	0,0507	0,0506	0,0500	0,0508	0,0527
$\hat{h}^2$	0,6345	0,2603	0,6296	0,6398	0,7166	0,6527	0,6034
$\hat{\sigma}_e^2/\hat{\sigma}_s^2$	5,304	14,366	5,353	5,252	4,582	5,129	5,7926

In table 3.3 the 95% Bayes credibility intervals for σ_e^2 , σ_s^2 , $h = \frac{4\sigma_s^2}{\sigma_s^2 + \sigma_e^2}$ and σ_e^2/σ_s^2 are given as well as the estimated means and medians of the posterior distributions. $E\left[\frac{\sigma_e^2}{\sigma_s^2} \mid \underline{y}\right]$ was calculated from (3.15). The skewness of the posterior distributions is an indication that the medians will most probably be more efficient estimates than the means.

TABLE 3.3: 95% Bayes credibility intervals for the variance components obtained from the Gibbs sampler

	σ_e^2	σ_s^2	$\frac{4\sigma_s^2}{\sigma_s^2 + \sigma_e^2}$	$\frac{\sigma_s^2}{\sigma_e^2}$
95% Intervals	0,0331– 0,0940	0,0012– 0,1554	0,0785– 2,8665	0,3695– 49,848
Estimated Medians	0,0527	0,0090	0,6034	5,7926
Estimated Means	0,0551	0,0245	0,8341	10,6734

A Bayesian approach to variance-component estimation has several practical advantages over a classical approach. Firstly the estimate for a variance is always positive and an interval estimate such as a highest posterior density region will not include negative values. This is in contrast to the classical estimators. Although ML and REML estimates are defined inside the appropriate parameter space, their asymptotic distributions (normal) can generate interval estimates that include negative values. This potentially embarrassing phenomenon is often overlooked in discussions of likelihood-based methods. Secondly, highest posterior density regions are never empty, whereas confidence intervals for the ratio of two variances can be empty. Finally, one can report the whole of the posterior probability distribution, not just a single number, and report some measure of posterior precision. Classical estimators generally have intractable sampling distributions and standard errors are hard to calculate.

Posterior distributions for the random (sire) effects can be obtained directly from the Gibbs sampler. The marginal posterior distributions for the elements of the vector $\underline{y} = [s_{11} \ s_{12} \ s_{13} \ s_{21} \ s_{22}]'$ are displayed in figure 3.3 and the estimates (mean values) are given in table 3.4. Only the means are tabulated since the distributions are more or less symmetrical.

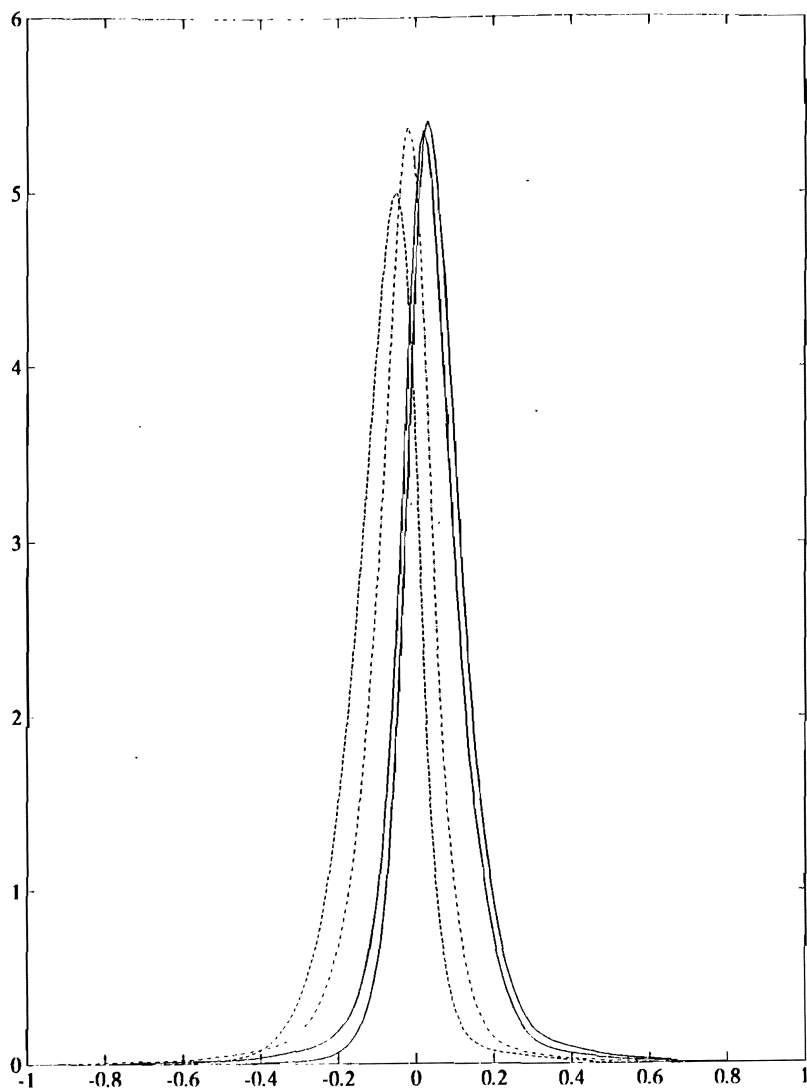


FIGURE 3.3: Posterior marginal for random effects u

TABLE 3.4: Solution for the random effect by various methods

	Method 3	ML	REML	MINQUE	MIQUEO	MINQUED 1	Gibbs
$\hat{s}_{11}$	0,059	0,063	0,059	0,060	0,063	0,060	0,063
$\hat{s}_{12}$	-0,092	-0,053	-0,091	-0,092	-0,097	-0,093	-0,082
$\hat{s}_{13}$	0,032	0,017	0,032	0,032	0,034	0,033	0,038
$\hat{s}_{21}$	-0,039	-0,023	-0,039	-0,039	-0,041	-0,040	-0,040
$\hat{s}_{22}$	0,039	0,023	0,039	0,039	0,041	0,040	0,035

In the case of fixed effects one is usually interested in differences between the effects. It might be of importance to investigate the difference among the age groups or between the two lines. To examine, for example, the difference between the two lines we need only to add the conditional distribution

$$(3.17) \quad (g_1 - g_2) | \underline{y}, \underline{\beta}_1, \underline{\beta}_2, \underline{u}, \sigma_e^2, \sigma_s^2 \sim N(\underline{\ell}'(X_3'X_3)^{-1}X_3'\underline{\theta}_{\beta_3}, \underline{\ell}'(X_3'X_3)^{-1}\underline{\ell}\sigma_e^2)$$

to the iterative procedure. Here $\underline{\ell} = [1, -1]'$ and $\underline{\theta}_{\beta_3}$ is defined in (3.10).

The density functions for the differences are given in figures 3.4 and 3.5. From the graphs it is clear that there are only small differences among the age groups or between the two lines with respect to average daily gain. In table 3.5 the estimated differences (posterior means) are compared with the classical estimates.

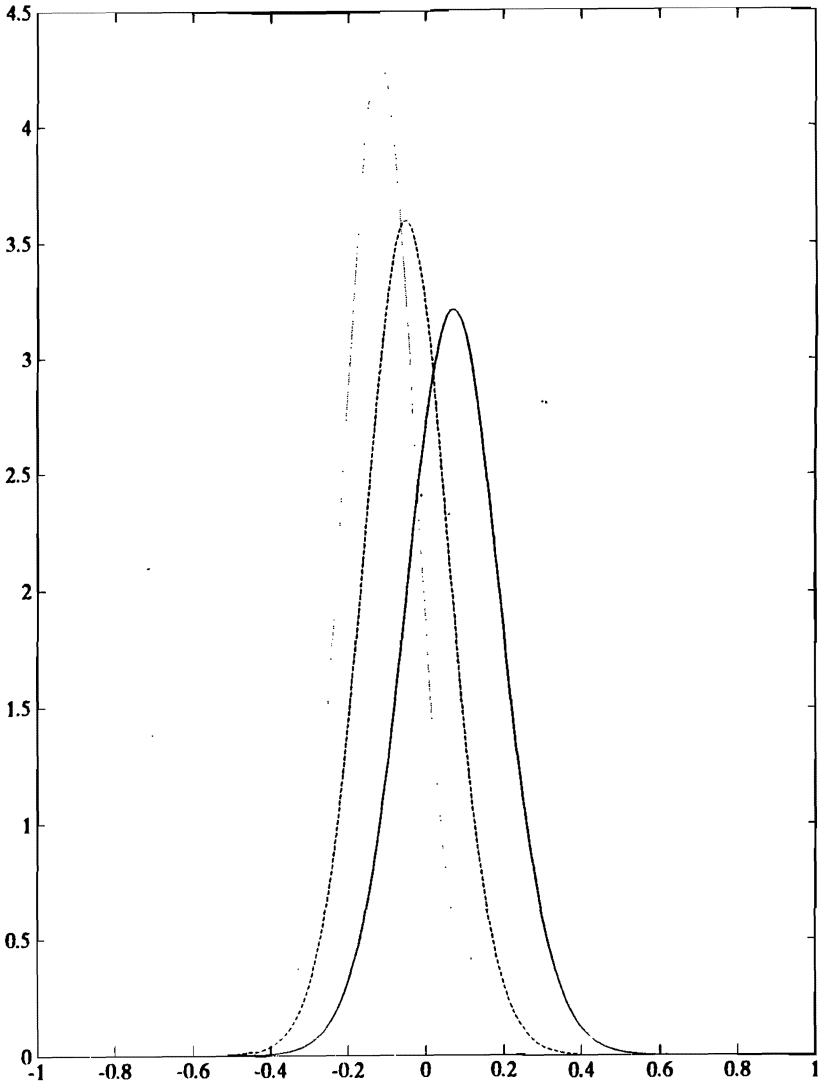


FIGURE 3.4: Posterior marginals for differences between age groups

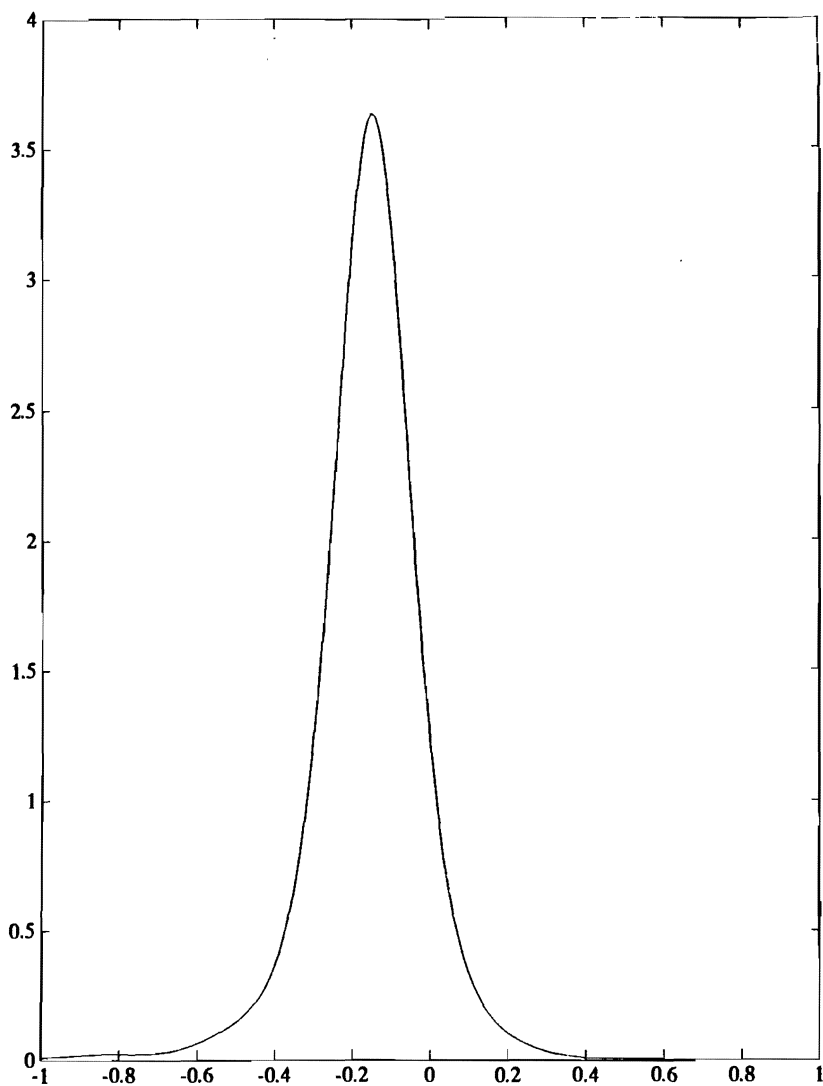


FIGURE 3.5: Posterior marginal for the difference between lines

TABLE 3.5: Differences for fixed effects

	Method 3	ML	REML	MINQUE	MIQUEO	MINQUED 1	Gibbs
$\hat{a}_1 - \hat{a}_2$	0,036	0,072	0,063	0,063	0,062	0,062	0,069
$\hat{a}_1 - \hat{a}_3$	-0,060	-0,048	-0,060	-0,060	0,062	-0,061	-0,052
$\hat{a}_2 - \hat{a}_3$	-0,123	-0,120	-0,123	-0,123	-0,124	-0,123	-0,121
$\hat{g}_1 - \hat{g}_2$	-0,148	-0,148	-0,148	-0,148	-0,148	-0,148	-0,158

4. NUMERICAL INTEGRATION TO OBTAIN MARGINAL INFERENCES ABOUT VARIANCE COMPONENTS

The joint unconditional posterior distribution of the variance components is

$$(4.1) \quad f(\sigma_s^2, \sigma_e^2 | \underline{y}) \propto (\sigma_e^2)^{-(n-p-q+2)/2} (\sigma_s^2)^{-(q+2)/2} |W'W + \Sigma|^{-\frac{1}{2}} \\ \exp[-(\underline{y}'\underline{y} - \hat{\theta}'W'\underline{y})/2\sigma_s^2]$$

$$\text{where } \hat{\theta} = (W'W + \Sigma)^{-1} W'\underline{y},$$

$W = [XZ]$ and X has full-column rank.

$$\text{Also } \Sigma = \begin{bmatrix} 0 & 0 \\ 0 & \sigma_e^2/\sigma_s^2 \end{bmatrix}$$

Gianola and Foulley (1990) derived equation (4.1) by integrating $\beta_1, \beta_2, \beta_3$ and $\underline{u}$ out of equation (3.5). By using numerical integration, the marginal posterior distribution of σ_e^2 and σ_s^2 can be obtained.

In figure 4.1 a comparison is made between the marginal posterior distributions of σ_e^2 using numerical integration and the distribution of σ_e^2 obtained from the Gibbs sampler.

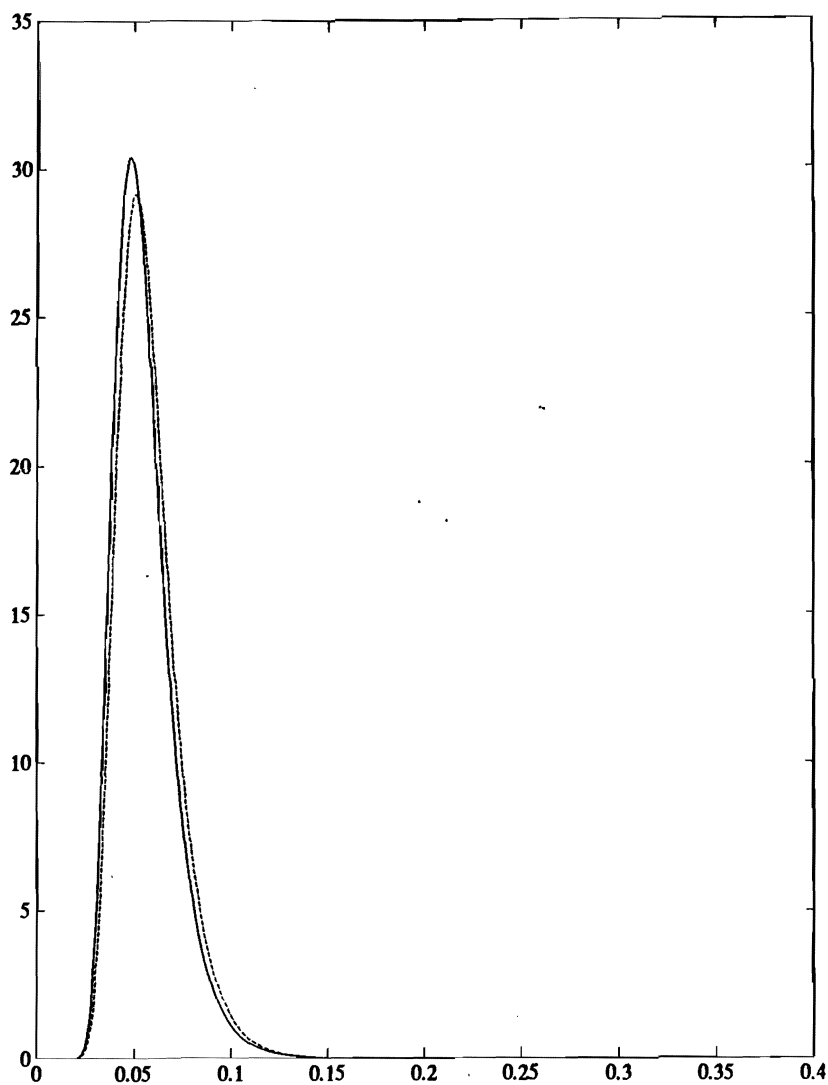


FIGURE 4.1: Posterior marginals for σ_{ϵ}^2

— Gibbs sampling Numerical integration

From the figure it is clear that the two densities are very similar. The central values and credibility intervals of the two distributions are for all practical purposes the same.

In table 4.1 the medians are compared.

TABLE 4.1 Comparison of the medians of the variance components using numerical integration and Gibbs sampling

	Numerical integration	Gibbs sampling
σ_e^2	0,0531	0,0527
σ_s^2	0,0093	0,0090

5. SENSITIVITY ANALYSIS

The convergence of the Gibbs algorithm is discussed in Gelfand and Smith (1990). Various techniques have been developed to check the convergence of the marginal distributions. This remains an area of active research.

In this section posterior densities, central values and credibility intervals are compared for various values of k . This comparison will also be made for an alternative Markov Chain Monte Carlo (MCMC) procedure in which after a long run (i.e. k sufficiently large) successive albeit correlated values of the random variable of interest are used to describe the unknown marginal distribution.

In figure 5.1 the marginal posterior distributions of σ_s^2 are compared for $k = 20, 30$ and 40 iterations and in figure 5.2 a similar comparison is made for σ_e^2 .

From the figures it is clear that the comparative densities are quite similar and that convergence has been achieved in all three cases, i.e. $k = 20$ and $m = 1\,000$ seems to be appropriate for this problem.

In figure 5.3 the marginal posterior density of σ_s^2 is illustrated for the alternative MCMC procedure with a "burn-in period" of $k = 500$ iteratives.

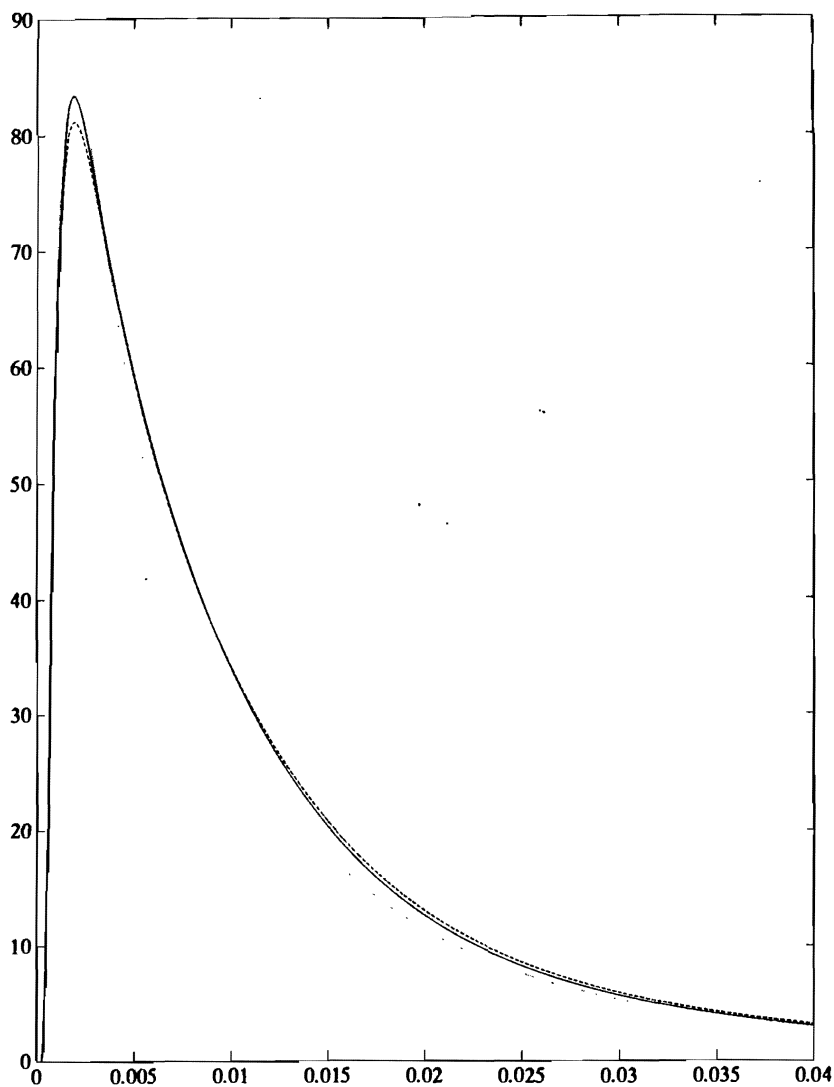


FIGURE 5.1: Posterior marginals of σ_ϵ^2 for various k
 $k = 20$ ——— $k = 30$ - - - $k = 40$

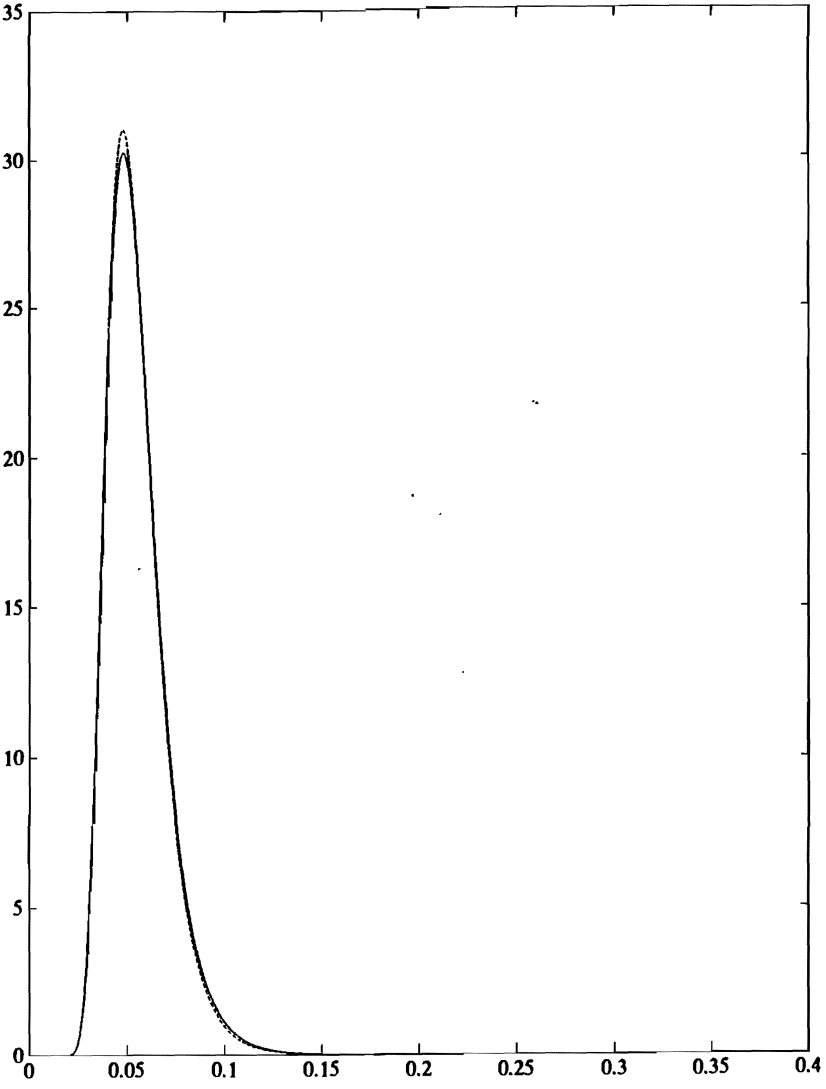


FIGURE 5.2: Posterior marginals for σ_{ϵ}^2 for various k
 $k = 20$ — $k = 30$ - - - $k = 40$

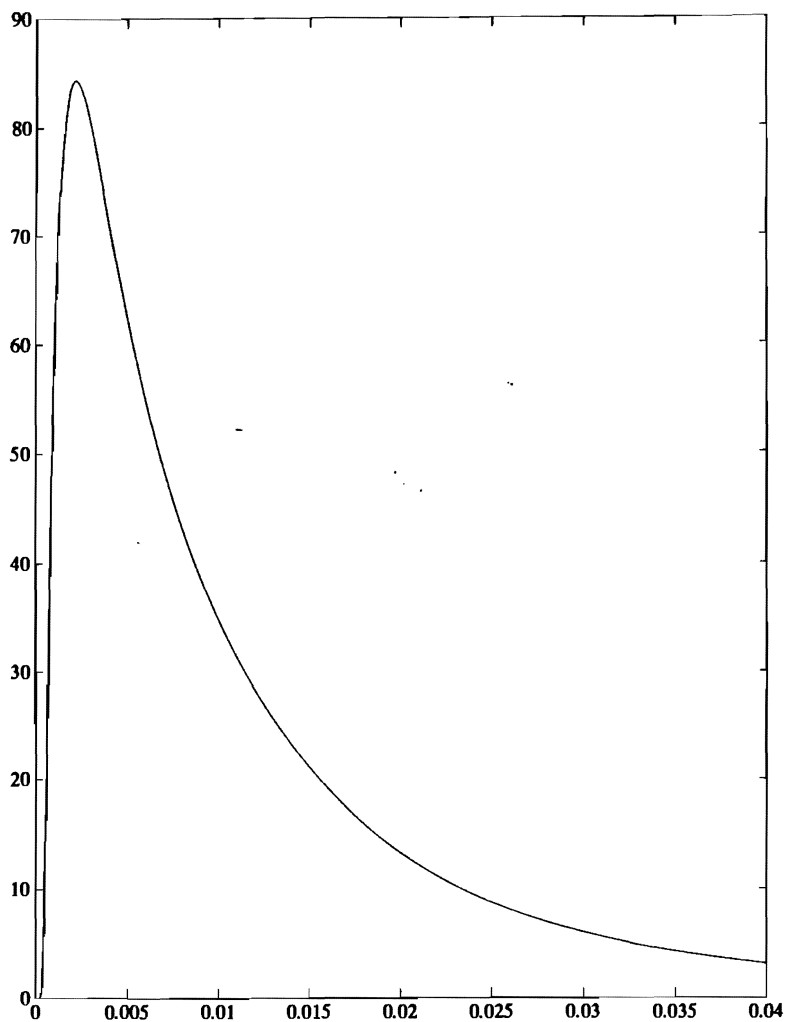


FIGURE 5.3: Posterior marginal for σ_s^2 for the MCMC procedure
 $k = 500$

A graphical comparison of the posterior densities of σ_s^2 (figures 5.1 and 5.3) suggests that they are quite similar. This similarity also exists for the other parameters of interest. In table 5.1 a comparison is made between the fixed effects (age of dam and breeding group) for various values of k .

TABLE 5.1 Differences of fixed effects for various k

	$k = 20$	$k = 30$	$k = 40$	$k = 500$
$\hat{a}_1 - \hat{a}_2$	0,069	0,072	0,073	0,072
$\hat{a}_1 - \hat{a}_3$	-0,052	-0,049	-0,047	-0,048
$\hat{a}_2 - \hat{a}_3$	-0,121	-0,121	-0,120	-0,120
$\hat{g}_1 - \hat{g}_2$	-0,158	-0,156	-0,152	-0,124

From the figures and table 5.1 it seems certain that the MCMC approach is applicable here. In this case only $(1\,000 + k) \times 6$ generations and substitutions are required. For $k = 500$ the number of generations is only 9 000. In the case of the usual procedure the number of generations is $20 \times 6 \times 1\,000 = 120\,000$. In this sense the Gibbs sampling procedure is computationally expensive.

Gelman and Rubin (1991), however, warned "that particularly during the first tentative examination of a new problem, it can be argued that monitoring the evolutionary behaviour of several runs of the chain starting from a wide range of initial values is necessary". It is for this reason that we have decided to stick to the "traditional" procedure.

For illustrative purposes a Bayesian analysis based on the "flat" prior specification $(p(\sigma_e^2, \sigma_s^2) \propto \text{constant})$ is also provided. This prior differs considerably from the improper prior $p(\sigma_e^2, \sigma_s^2) \propto \sigma_e^{-2} \sigma_s^{-2}$ defined in (3.4). The latter was suggested by Hill (1965), which is a naive two-dimensional extension of the familiar non-informative prior for a variance. For further details see Gelfand *et al.* (1990). In figures 5.4 and 5.5 the marginal posterior densities of σ_s^2 and σ_e^2 are displayed for the "flat" prior specifications.

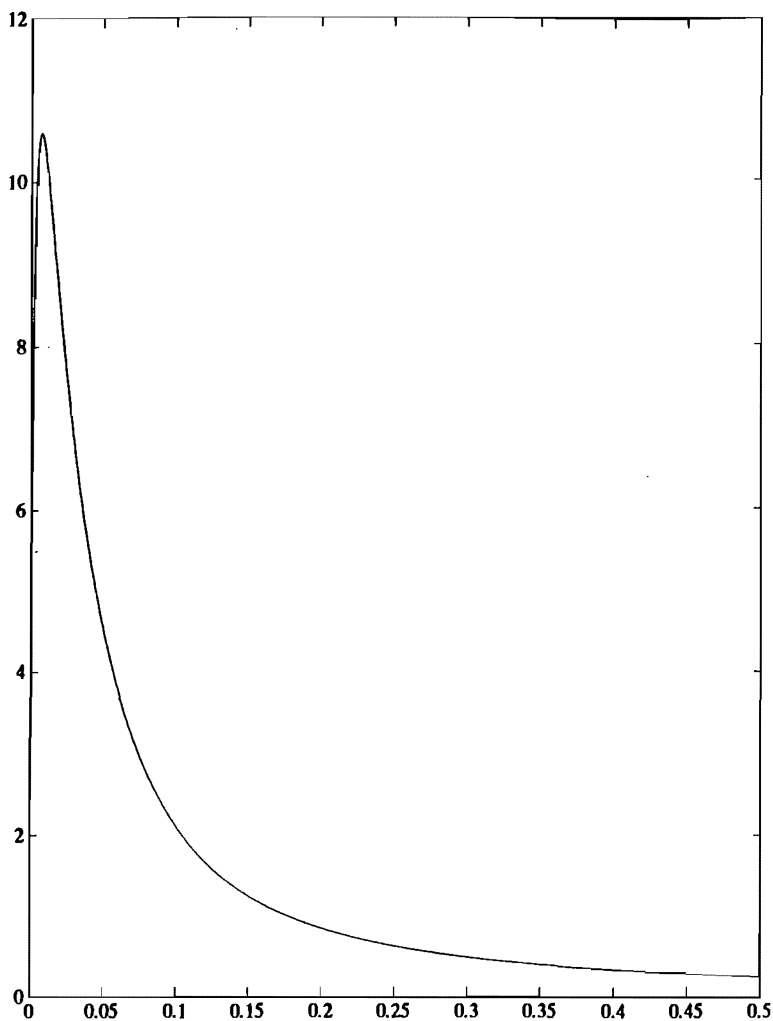


FIGURE 5.4: Posterior marginal for σ_ϵ^2 based on $p(\sigma_\epsilon^2) \propto \text{constant}$

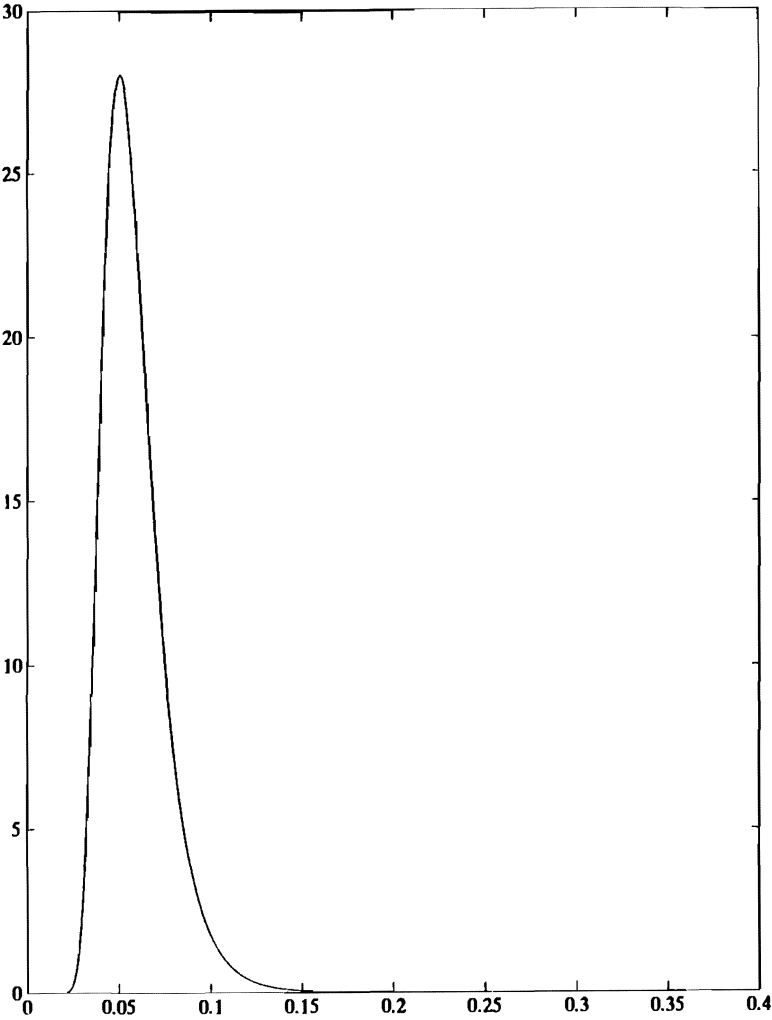


FIGURE 5.5: Posterior marginal for σ_{ϵ}^2 based on $p(\sigma_{\epsilon}^2) \propto \text{constant}$

Comparing figures 5.1 and 5.4 shows that there is a substantial difference in the marginal posterior densities of σ_s^2 . This is not surprising since the marginal conditional posterior distributions of σ_s (based on the two prior specifications) differ quite a lot. This is especially true if q is small (see equation (3.1)). For our problem $q = 5$.

This difference does not occur in the posterior distributions of the other parameter values. This can be observed from figures 5.2, 5.3 and 5.5 where the marginal posterior distributions of σ_e^2 are displayed and also from table 5.2 where central values and credibility intervals of σ_e^2 are given. The reason for the greater similarity in the posterior distribution of σ_e^2 is that the degrees of freedom "within groups" is greater than the degrees of freedom "between groups". So it could be expected that there is more information in the data about σ_e^2 than about σ_s^2 .

Chaloner (1987) considered also the following improper non-informative prior distributions $p(\sigma_e^2, \sigma_s^2) \propto (\sigma_e^2 + \sigma_s^2)^{-1}$, $p(\sigma_e^2, \sigma_s^2) \propto (\sigma_e^2)^{-1}(\sigma_e^2 + \sigma_s^2)^{-1}$ and $p(\sigma_e^2, \sigma_s^2) \propto (\sigma_e^2)^{-\frac{1}{2}}(\sigma_e^2 + \sigma_s^2)^{-\frac{3}{2}}$ but because they gave similar results they are not included in our study.

DISCUSSION

Harville (1990) stated: "A more extensive use of Bayesian ideas by animal breeders and other practitioners is desirable and is more feasible from a computational standpoint than commonly thought. The Bayesian approach can be used to devise prediction procedures that are more sensible – from both a Bayesian and frequentist perspective – than those in current use."

In this paper we have used the Bayesian method of variance component estimation in mixed linear models based on the principle of finding the marginal posterior distributions of the unknown parameters. We showed that an adaptive Monte Carlo integration technique known as the Gibbs sampler provides a simple and efficient framework for calculating these distributions. Although other techniques for finding posterior marginals are

Table 5.2: Central values and credibility intervals for σ_e^2 based on two prior specifications for various k

	Improper Prior $p(\sigma_e^2, \sigma_s^2) \propto \sigma_e^{-2} \sigma_s^{-2}$			Flat prior $p(\sigma_e^2, \sigma_s^2) \propto \text{constant}$
	$k = 20$	$k = 30$	$k = 40$	$k = 20$
Posterior Mean	0,0551	0,0546	0,0549	0,0583
Posterior Median	0,0527	0,0517	0,0525	0,0589
95% Credibility Interval	0,0331–0,0940	0,0325–0,0891	0,0318–0,0898	0,0327–0,1071

available. The Gibbs sampler has proved very efficient in our example, converging remarkably quickly. Indeed we need not compromise high dimensional integration by replacing integration with estimation as has often been done in practice (see for example Gianola and Fernando (1986) and Gianola and Foulley (1990)). The Gibbs sampling approach is notable for its ease of implementation to a wide variety of models. The conceptual simplicity of the approach makes it easy to write the necessary programs and having written the programs, it is easy to change them to handle new problems. Competitive results can therefore be obtained without sophisticated numerical techniques and very little effort. The ultimate value of the Gibbs sampler therefore lies in its practical potential.

ACKNOWLEDGEMENTS

The authors are grateful to the two referees, the editor and associate editor for valuable comments and suggestions. We also wish to thank Mrs F. Merriman for the typing of the original manuscript.

REFERENCES

- ACHCAR, J.A. & SMITH, A.F.M. (1990). *Aspects of reparameterization in approximate Bayesian inference*. In *Essays in honour of George A. Barnard*, J. Hodges Editor, North Holland.
- BOX, G.E.P. & TIAO, G.C. (1973). *Bayesian inference in statistical analysis*. Addison-Wesley, Redding, MA.
- CASELLA, G. & GEORGE, E.I. (1990). An introduction to Gibbs sampling. *Technical Report No. 96*. The University of Chicago, Graduate School of Business.
- CHALONER, K. (1987). A Bayesian approach to the estimation of variance components for the unbalanced one-way random model. *Technometrics*, 29, 323-337.

- GELFAND, A.E. & SMITH, A.F.M. (1990). Sampling based approaches to calculating marginal densities. *J. Am. Statist. Assoc.*, **85**, 972-985.
- GELFAND, A.E. & SMITH, A.F.M. (1991). Gibbs sampling for marginal posterior expectations. *Communications in Statistics, Theory and Methods*, **20** (5 and 6), 1747-1766.
- GELFAND, A.E. HILLS, S.E., RACINE-POON, A., SMITH, A.F.M. (1990). Illustration of Bayesian inference in normal data models using Gibbs sampling. *J. Am. Statist. Assoc.*, **85**, 972-985.
- GELMAN, A. & RUBIN, D.R. (1992). A single series from the Gibbs sample provides a false sense of security. In *Bayesian Statistics*, **4** (editors J.J. Bernardo, J. Berger, A.P. Dawid and A.F.M. Smith), 627-633, Oxford University Press.
- GEMAN, S. & GEMAN, D. (1984). Stochastic relaxation, Gibbs distributions and the Bayesian restoration of images. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, **6**, 721-741.
- GEWEKE, J. (1988). Antithetic acceleration of Monte-Carlo integration in Bayesian inference. *Journal of Econometrics*, **38**, 73-90.
- GIANOLA, D. & FERNANDO, R.L. (1986). Bayesian methods in animal breeding theory. *Journal of Animal Science*, **63**, 217-244.
- GIANOLA, D. & FOULLEY, J.L. (1990). Variance estimation from integrated likelihoods (VEIL). *Journal of Genetic Selection and Evolution*, **22**, 403-417.
- HARVEY, W.R. (1975). Least-squares analysis of data with unequal subclass numbers. Agriculture Research Service, United States Department of Agriculture ARS H-4.
- HARVILLE, D.A. (1977). Maximum likelihood approach to variance component estimation and to related problems. *J. Am. Statist. Assoc.*, **72**, 320-340.
- HARVILLE, D.A. (1990). BLUP (Best linear unbiased prediction) and beyond. In *Advances in Statistical Methods for Genetic Improvement of Livestock* (Gianola, D., Hammond, K. Eds.), Springer-Verlag, New York-Heidelberg-Berlin, 239-276.
- HENDERSON, C.R. (1984). Applications of linear models in animal breeding. University of Guelph, Guelph, Ontario, Canada.

- HILL, G.M. (1965). Inference about variance components in the one-way model. *J. Am. Statist. Assoc.*, **60**, 806-825.
- HILL, B.M. (1977). Exact and approximate Bayesian solutions for inference about variance components and multivariate inadmissibility; in *New Developments in the Applications of Bayesian Methods*, Eds. A. Ayka and C. Brumat, Amsterdam: North Holland, 129-152.
- KENNEDY, B.W. (1981). Variance component estimation and prediction of breeding values. *Canadian Journal of Genetics and Cytology*, **XXIII**, No. 4, 565-578.
- LINDLEY, D.V. & SMITH, A.F.M. (1972). Bayes estimates for the linear model. *J. Roy. Statist. Soc., Ser. B*, **34**, 1-41.
- NAYLOR, J.C. & SMITH, A.F.M. (1982). Applications of a method for the efficient computation of posterior distributions. *Applied Statistics*, **31**, 214-225.
- NAYLOR, J.C. & SMITH, A.F.M. (1988). Econometric illustrations of novel numerical integration strategies for Bayesian inference. *Journal of Econometrics*, **38**, 103-126.
- RUBIN, D.B. (1981). Estimation in parallel randomized experiments. *Journal of Educational Statistics*, **6**, 377-400.
- SEARLE, S.R. (1971). *Linear models*. John Wiley and Sons, New York.
- SEARLE, S.R. (1979). Notes on variance component estimation : a detailed account of maximum likelihood and kindred methodology. *Biometrics Unit.*, Cornell Univ., Ithaca, NY.
- SKENE, A.M. & SHAW, J.E.H. (1983). Computing marginal distributions for the dispersion parameters of analysis of variance models. *The Statistician*, **32**, 99-108.
- SMITH, A.F.M., SKENE, A.M., SHAW, J.E.H., NAYLOR, J.C. & DRANSFIELD, M. (1985). The implementation of the Bayesian paradigm, *Communications in Statistics, Part A - Theory and Methods*, **14**, 1079-1102.
- SMITH, A.F.M., SKENE, A.M., SHAW, J.E.N., & NAYLOR, J.C. (1987). Progress with numerical and graphical methods for Bayesian statistics. *The Statistician*, **36**, 75-82.
- TIERNEY, L. & KADANE, J.B. (1986). Accurate approximations for posterior moments and marginal densities. *J. Am. Statist. Assoc.*, **81**, 82-86.
- ZELLNER, A. (1971). *An introduction to Bayesian inference in econometrics*. J. Wiley & Sons, New York.